

CITYTWIN · DESIGN DECISION RECORD

CityTwin TRL 2 review decisions

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review recommendations adopted for TRL 3 work and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** proposed. The recommendations in items D1 to D11 are adopted for TRL 3 work pending Amish's review; item O1 remains "Proposed, awaiting Amish".

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed ten items as "Proposed, awaiting Amish", and the design precis (CTW-PRC-001 v0.2, Key design choices) gave a recommendation for most of them. On 2026-09-25 Amish asked for this batch of repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He has not reviewed this repo's items one by one. Under that instruction, every item that carries a recommendation is adopted as recommended for TRL 3 work, open for his review. Items without a recommendation stay open. A recommended change to budget_usd is not applied: the figure is recorded here and in docs/REVIEW.md as awaiting Amish.

■ Options considered

The options for each item are those listed in docs/REVIEW.md (session 2026-09-25, /populate) and in CTW-PRC-001 v0.2, Key design choices. D11 is an interface proposal made in this session from the sibling repos' TRL 3 reviews.

■ Decision

Table 1. Items adopted for TRL 3 work.

#	Item	Adopted choice	Status
D1	Budget	Option (b): the budget covers the kiosk only (items 1 to 12, 14 and 15); the TwinKit gateway is costed in the TwinKit repo. The recommended new figure of about \$750 is not applied: <code>budget_usd</code> stays at \$600 and \$750 is recorded as awaiting Amish. R16 is stated against both figures; R17 is redefined as a reported figure, not a budget target.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review (new <code>budget_usd</code> figure awaiting Amish)
D2	Display	13.3 in color e-paper for the street kiosk, with an indoor LCD variant for libraries and city hall lobbies (variant not modeled or costed at TRL 3).	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D3	Climate range	Sheltered or indoor siting for the pilot, then a heater study at TRL 3. R12 is redefined as the pilot range (sheltered site, air 0 to 40 °C, no direct sun on the window); the outdoor range of -20 to +45 °C moves to a new R20 as the target after the pilot. The heater study is CTW-CAL-001 section F.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D4	Gateway location	TwinKit gateway in the kiosk cabinet for the pilot, on the upper DIN rail laid out as in TWK-DWG-001, with a coax lead to the post-top antenna; rooftop gateway for a full neighborhood.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D5	Power	Mains, with a solar head-only variant for sites without power (variant not modeled or costed at TRL 3).	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D6	Privacy settings	Small-count threshold of 5 per hour; raw records kept two years; hourly aggregates kept indefinitely (sized for 10 years in CTW-CAL-001).	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D7	Open data license	CC BY 4.0.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D8	Controls	Three push buttons rather than a touch screen, with the QR code handing richer use to the visitor's phone.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D9	Reference neighborhood	The 46-node reference neighborhood is kept for sizing. Reporting intervals now follow the sibling repos' TRL 3 figures (LoadZone 124 uplinks a day, not 64); CrossSafe's hourly summary remains a CityTwin assumption.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D10	Architecture	One-way publishing (no inbound connections to the gateway) and privacy rules enforced at the gateway as well as in the nodes (CTW-PRC-001 v0.2, Key design choices).	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review

#	Item	Adopted choice	Status
D11	Upload paths for PotholeLog and DockHub (new, from the sibling TRL 3 reviews)	The gateway pulls, by outbound HTTPS, the fleet operator's published daily PotholeLog segment files and DockHub's per-dock hourly aggregates from its back end. This keeps D10, because the gateway still accepts no inbound connection. To be agreed with both repos; neither was edited.	Proposed in this session; used for TRL 3 sizing, awaiting Amish

Table 2. Items that remain open.

#	Item	Status
O1	First partner city or neighborhood for co-design and a pilot site. No recommendation was made.	Proposed, awaiting Amish

No reworded pitch or problem line was recommended, so `project.yaml` and `README.md` keep the existing wording.

■ Consequences

- `project.yaml` : only the TRL fields and evidence change. `budget_usd` stays at \$600.
- CTW-PRB-001, CTW-PRC-001 and CTW-REQ-001 are revised to v0.3. The design choices in D1 to D10 are no longer described as "proposed" but as adopted for TRL 3 pending Amish's review. R12 and R17 are redefined, R20 is added and R1 is restated for the pull paths (D11).
- CTW-CAL-001 finds R10 (button response), R16 (kiosk cost against \$600), R18 (backup in the worst case) and R20 (street climate) not met, and R1 and R12 at risk. New proposals from those findings are listed in `docs/REVIEW.md` and are not applied.
- Cross-repo: the gateway cost is taken from TwinKit's TRL 3 BOM (\$290.00), matching TwinKit's own O1. CrossSafe defines no LoRaWAN uplink (its radio runs LoRa point to point between the two sides), so CityTwin's CrossSafe layer depends on an interface that CrossSafe has not yet defined. LoadZone's sign option needs class C downlinks from the TwinKit network server. See `docs/REVIEW.md` .
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building or testing.